

Handball Data Analytics Project - Bundesliga 2023

ASK

Questions to solve:

- Who were the top scorers?
- Who had the most assists?(best playmaker)
- Who had the most blocks and steals?(best defender)
- Who had the most 2 minutes and red cards?
- Who had the best shooting percentage?
- Who had the best hpi?
- Which team had the most shooters in the top 50?
- What is the correlation between effectiveness and hpi?
- efficiency vs usage: which players have >65%, more than 30 assists and less than 15 technical errors?
- Which players have the biggest sum of steals and blocks compared the 2 minute suspensions?
- Are the teams with the most players in the top 50 hpi in the top of the championship?

PREPARE

Data source: this data contains handball statistics from professional handball player that play in the top division of Germany, including:

- **NAME** : Name of a player (can be used as ID), string
- **CLUB** : Club in which player is currently playing (18 total clubs), categorical
- **POSITION** : Players position (7 total positions), categorical
- **P** : Number of games player played this season, numerical
- **G** : Number of goals player scored this season, numerical
- **M** : Number of shots player missed this season, numerical
- **FG** : Number of field goals player scored this season (G - G.1), numerical
- **G.1** : Number of penalty kicks player scored this season (G - FG), numerical
- **%** : Players shot percentage this season ($G/(G+M)*100$), string
- **AS** : Number of assists player had this season, numerical
- **TF** : Number of technical fouls player had this season, numerical
- **ST** : Number of steals player had this season, numerical
- **BL** : Number of shots player has blocked this season, numerical
- **YC** : Number of yellow cards player received this season, numerical
- **2MIN** : Number of two minutes suspension player received this season (usually for hard fouls), numerical
- **RC** : Number of red cards player received this season (can be received direct or after getting 2 minutes suspension third time in a game), numerical
- **BK** : Number of red cards player received this season (usually after very hard foul or making problems to referee), numerical
- **HPI** : Average handball performance index (sophisticated method of measuring how well players played game created by league engineers), numerical

Data Limitations: Small and non-diverse sample, collected in 2022-23 (older data).

Data Integrity Checks: Verified completeness and consistency of records, removed duplicate rows and users with missing activity logs, i renamed some columns for better understanding of the data.

clean

Tools Used: SQL and looker studio.

Data Cleaning Steps: For the preparation of the data I removed null and duplicate values. These steps ensured that the data would be reliable, clean and structured.

1. I created a table that is called ***handball_stats_bundesliga***, so that i can clean my data.
2. Removed duplicate data

```
SELECT *  
  
FROM (  
  
  select distinct NAME  
  
  from `lithe-aileron-433816-k5.handball.handball_stats_bundesliga`  
  
  qualify ROW_NUMBER() OVER (PARTITION BY NAME ORDER BY G DESC)=1  
  
)
```

3 Renamed columns

```
ALTER TABLE `lithe-aileron-433816-k5.handball.handball_stats_bundesliga`  
  
RENAME COLUMN `%` TO PERCENTAGE,  
  
RENAME COLUMN `AS` TO ASSISTS,  
  
RENAME COLUMN P TO GAMES_PLAYED,  
  
RENAME COLUMN G TO TOTAL_GOALS,  
  
RENAME COLUMN M TO GAMES_STARTED,  
  
RENAME COLUMN FG TO FIELD_GOALS,  
  
RENAME COLUMN G_1 TO GOALS_PER_GAME,  
  
RENAME COLUMN TF TO TURNOVERS,  
  
RENAME COLUMN ST TO STEALS,
```

```
RENAME COLUMN BL TO BLOCKS;
```

4. Converted types

In column `percentages` i run a function so it can have only 2 digits.

```
update
```

```
`handball.handball_stats_bundesliga`
```

```
set PERCENTAGE = round(PERCENTAGE, 2)
```

```
where PERCENTAGE is not null
```

Then

```
update
```

```
`handball.handball_stats_bundesliga`
```

```
set PERCENTAGE = cast(round(PERCENTAGE * 100, 0) as int64)
```

```
where PERCENTAGE is not null
```

5. I changed the name of goals_per_game to penalty_shots because it was the right name to put.

```
alter table `lithe-aileron-433816-k5.handball.handball_stats_bundesliga`
```

```
rename column GOALS_PER_GAME to
```

```
Penalty_shots;
```

6. I substituted all the null data (if existed) with zero

update

```
`handball.handball_stats_bundesliga`
```

SET

```
ASSISTS = IFNULL(ASSISTS, 0),  
TURNOVERS = IFNULL(TURNOVERS, 0),  
STEALS = IFNULL(STEALS, 0),  
BLOCKS = IFNULL(BLOCKS, 0),  
YC = IFNULL(YC, 0),  
`2MIN` = IFNULL(`2MIN`, 0),  
RC = IFNULL(RC, 0),  
PENALTY_SHOTS = IFNULL(PENALTY_SHOTS, 0)
```

WHERE true;

7. I searched for outliers with no data to display.

```
SELECT * FROM `handball.handball_stats_bundesliga`  
WHERE TOTAL_GOALS > 500 OR TOTAL_GOALS < 0;
```

8. Validated data

-Logic check with no data to display

```
SELECT NAME, TOTAL_GOALS, MISSED_SHOTS, FIELD_GOALS  
FROM `handball.handball_stats_bundesliga`  
WHERE TOTAL_GOALS < FIELD_GOALS -- this would be a mistake  
OR TOTAL_GOALS < 0;
```

- Null prices with no data to display

```
SELECT COUNT(*) as missing_values  
  
FROM `handball.handball_stats_bundesliga`  
  
WHERE PERCENTAGE IS NULL  
  
OR HPI IS NULL  
  
OR NAME IS NULL;
```

-Checked for range validation

```
SELECT  
  
MIN(PERCENTAGE) as min_perc,  
  
MAX(PERCENTAGE) as max_perc,  
  
MIN(TOTAL_GOALS) as min_goals,  
  
MAX(TOTAL_GOALS) as max_goals  
  
FROM `handball.handball_stats_bundesliga`;
```

- Uniqueness check with no data to display

```
SELECT NAME, COUNT(*)  
  
FROM `handball.handball_stats_bundesliga`  
  
GROUP BY NAME  
  
HAVING COUNT(*) > 1;
```

DATA ANALYSIS AND DATA VISUALIZATION

- Who were the top scorers?

SELECT

NAME,

TOTAL_GOALS,

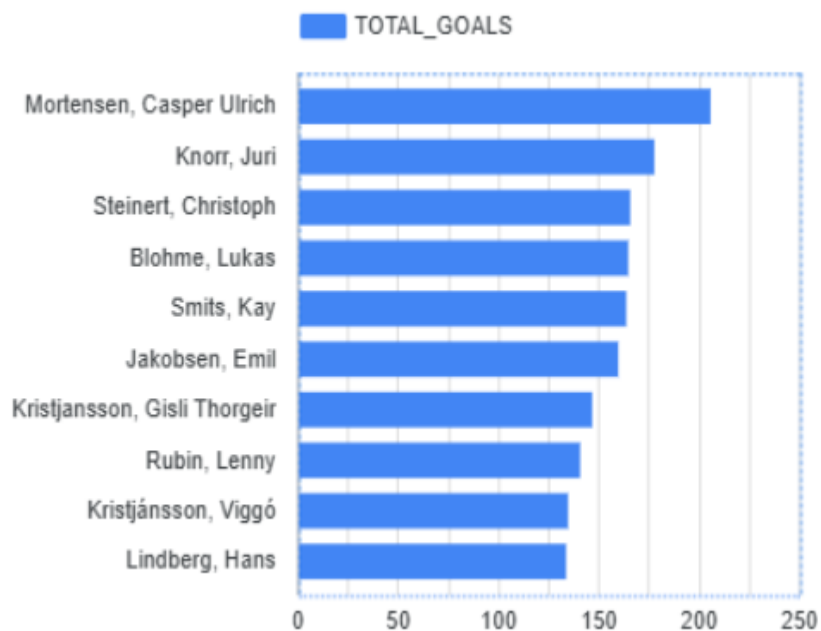
CLUB,

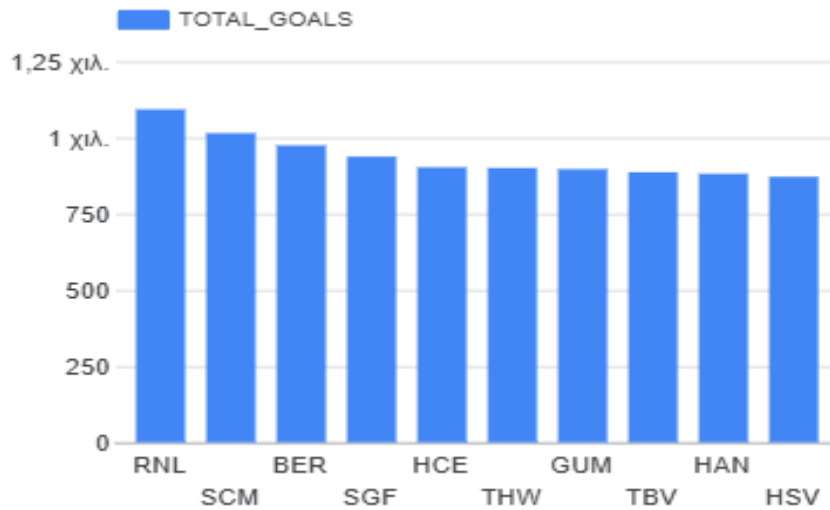
POSITION,

GAMES_PLAYED

FROM `lithe-aileron-433816-k5.handball.handball\_stats\_bundesliga`

ORDER BY TOTAL_GOALS DESC





Insight: This list identifies the league's most prolific offensive threats. Casper Ulrich Mortensen leads the scoring charts, showing high consistency as a Left Wing (LW), which often indicates a primary role in fast breaks and penalty taking.

- who had the most assists?(best playmaker)

SELECT

NAME,

ASSISTS,

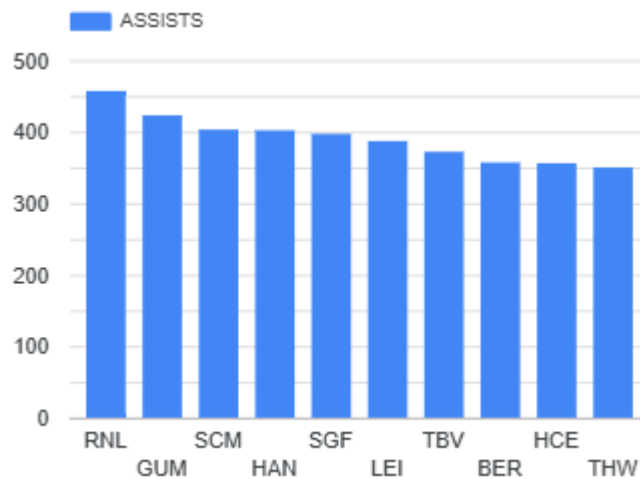
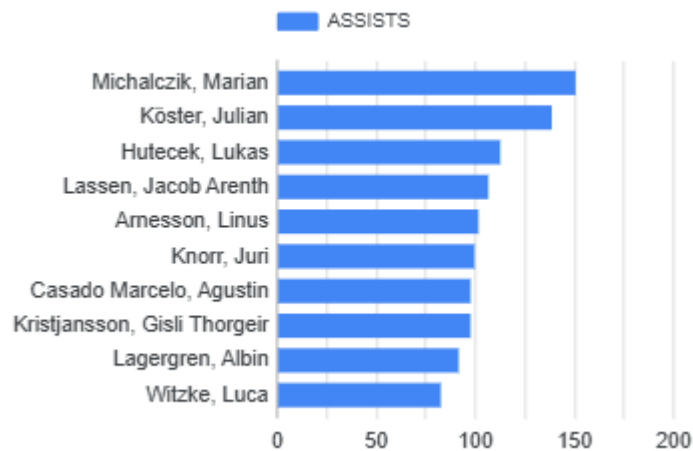
CLUB,

POSITION,

GAMES_PLAYED

FROM `lithe-aileron-433816-k5.handball.handball\_stats\_bundesliga`

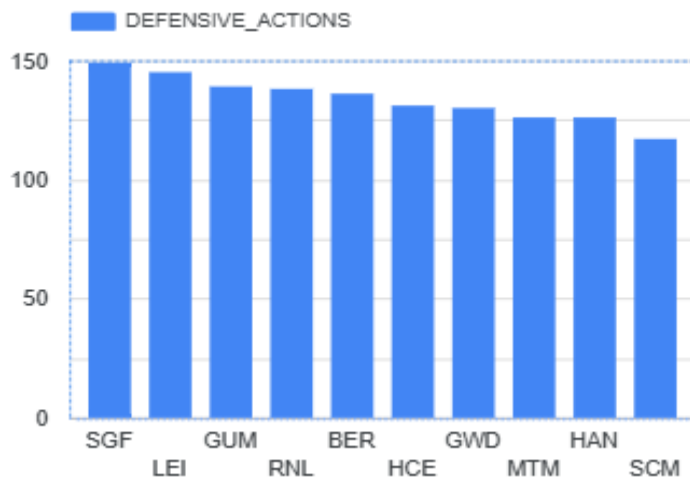
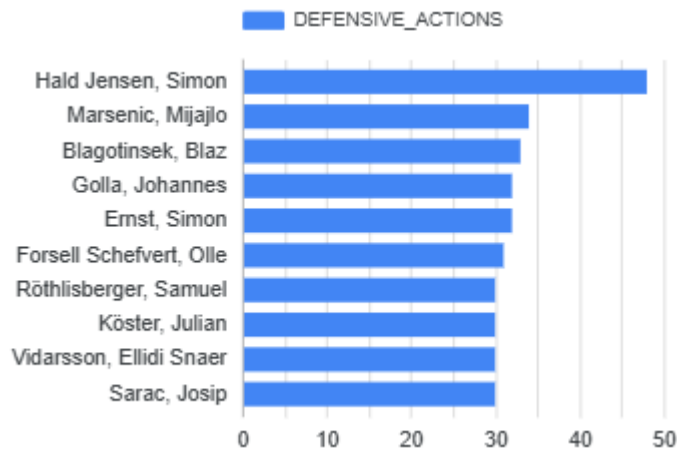
ORDER BY ASSISTS DESC



Insight: Assists are a key indicator of court vision and playmaking ability. The top players here are the organisers of their teams, responsible for creating high-quality scoring opportunities for their teammates.

- who had the most blocks and steals?(best defender)

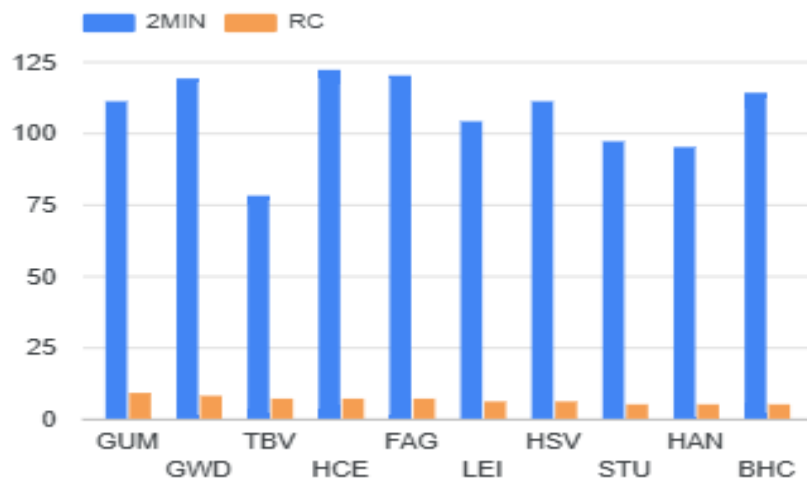
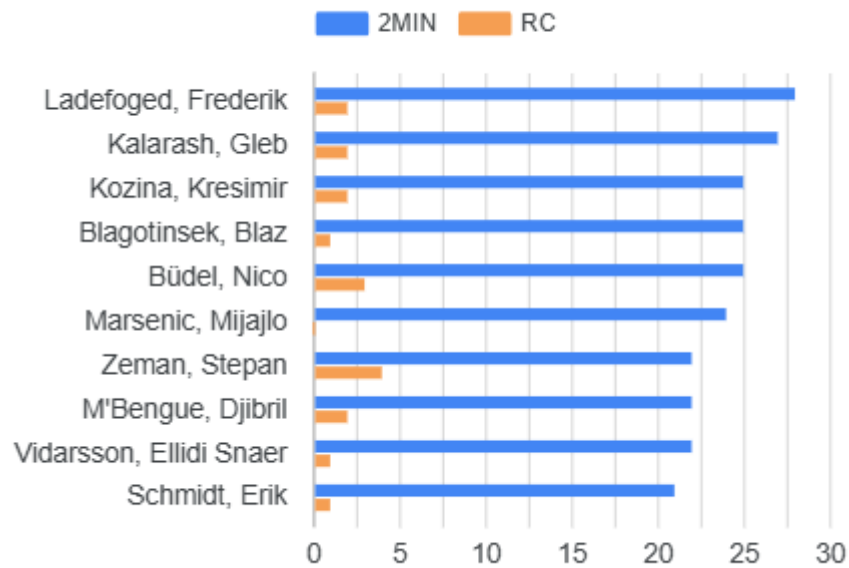
```
SELECT NAME, CLUB, (BLOCKS + STEALS) AS DEFENSIVE_ACTIONS
FROM `lithe-aileron-433816-k5.handball.handball_stats_bundesliga`
ORDER BY DEFENSIVE_ACTIONS DESC
```



Insight: This metric highlights defensive specialists. Players with high combined blocks and steals disrupt the opponent's rhythm and are crucial for transitioning from defense to fast-break attacks.

- who had the most 2 minutes and red cards?

```
SELECT NAME, CLUB, `2MIN`, RC
FROM `lithe-aileron-433816-k5.handball.handball_stats_bundesliga`
ORDER BY RC DESC, `2MIN` DESC
```



Insight: High disciplinary records often point to "aggressive" defensive roles. While these players are physically dominant, high numbers in this category can be a liability, highlighting the fine line between tough defense and disadvantageous penalties.

- who had the best shooting percentage?

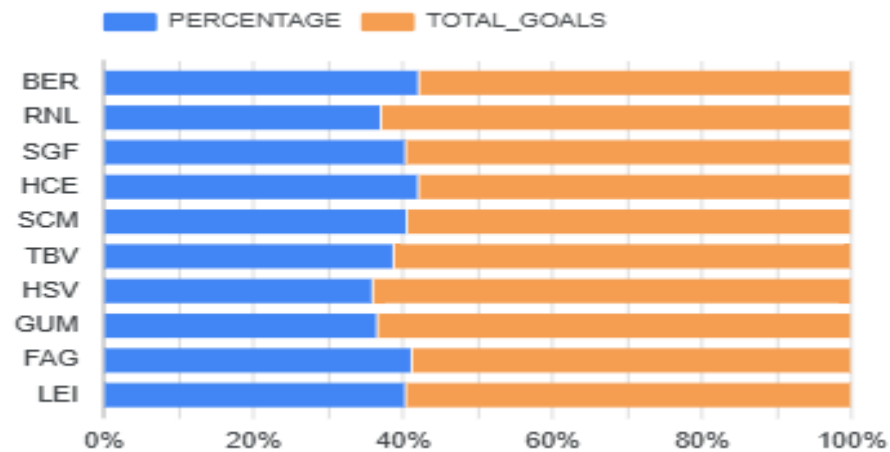
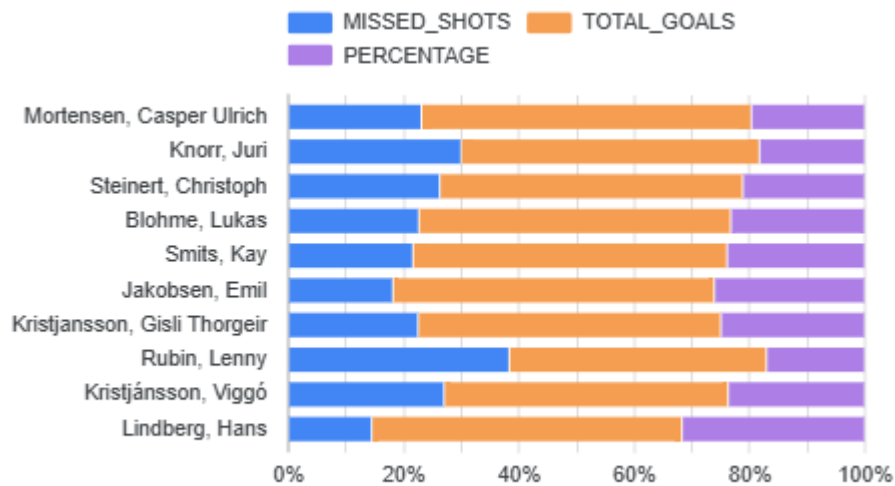
SELECT

NAME,

```

club,
PERCENTAGE,
MISSED_SHOTS,
TOTAL_GOALS
FROM `lithe-aileron-433816-k5.handball.handball_stats_bundesliga`
WHERE TOTAL_GOALS > 60
ORDER BY PERCENTAGE DESC;

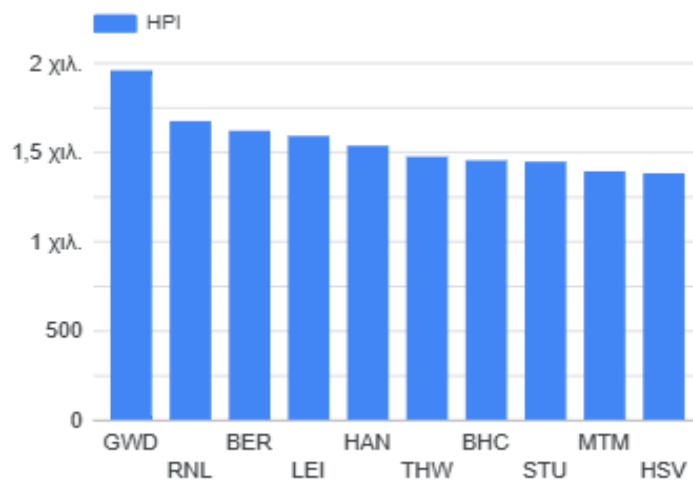
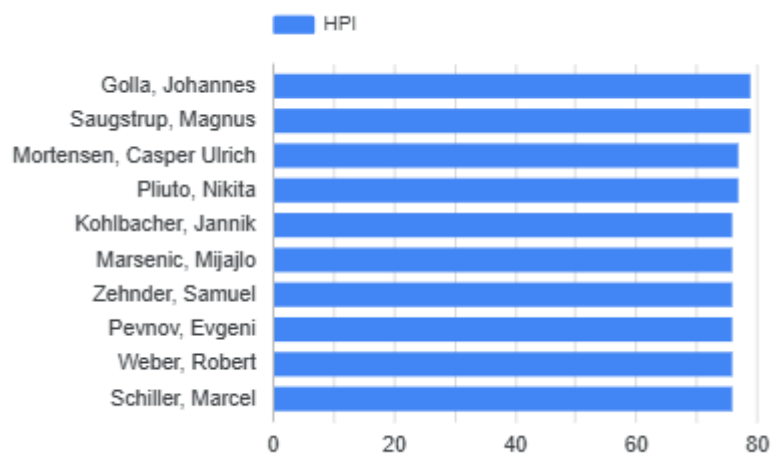
```



Insight: Efficiency is just as important as volume. By filtering for players with significant goal tallies, we identify the most clinical finishers who rarely waste possessions, often led by specialized wings and pivots.

- who had the best hpi?

```
SELECT
    NAME,
    club,
    HPI
FROM `lithe-aileron-433816-k5.handball.handball_stats_bundesliga`
ORDER BY HPI DESC;
```



Insight: The Handball Performance Index (HPI) is the ultimate holistic metric. The leaders in this category are the most complete players in the Bundesliga, contributing significantly across multiple phases of the game.

- which team had the most shooters in the top 50?

```
WITH Top50Players AS (
    SELECT
        NAME,
        CLUB
    FROM `lithe-aileron-433816-k5.handsball.handsball_stats_bundesliga`
    ORDER BY HPI DESC
    LIMIT 50
)
SELECT
    club,
    COUNT(*) AS player_count
FROM Top50Players
GROUP BY club
ORDER BY player_count DESC;
```

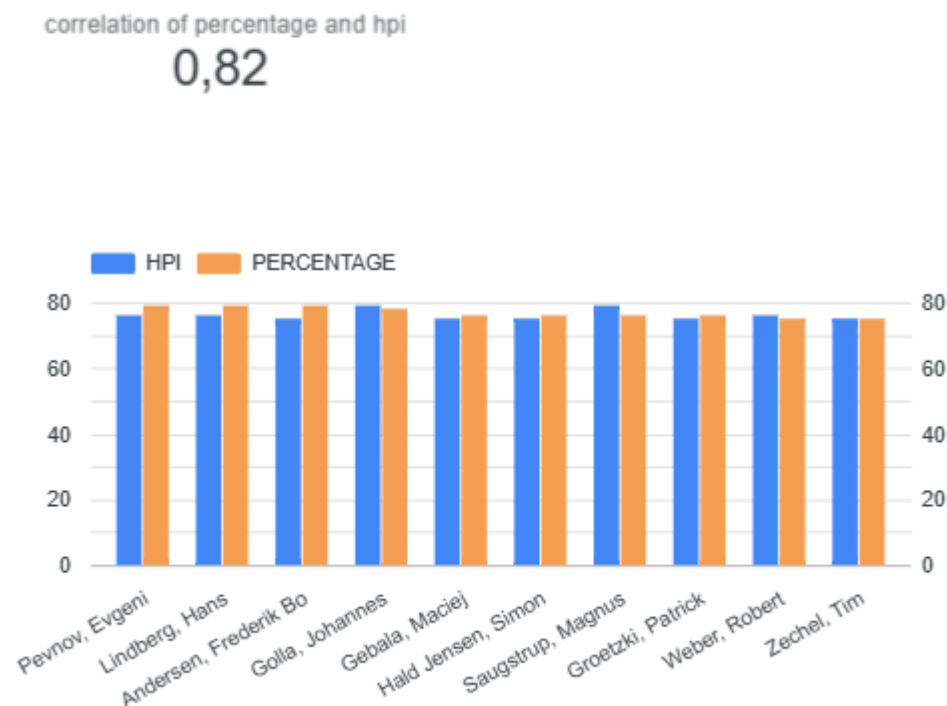
	club	average_hpi_o...	players_in_... ▾
1.	BER	75	6
2.	SGF	75	5
3.	HSV	74,4	5
4.	HAN	74	5
5.	SCM	75,5	4
6.	TBV	74,8	4
7.	GUM	74	3
8.	MTM	73,3	3
9.	THW	73,7	3
1...	HCE	74	2

Insight: This chart visualizes roster depth. Teams like SC Magdeburg and Füchse Berlin dominate this list, showing that elite teams invest in multiple high-performing assets rather than relying on a single "star" player.

- What is the correlation between effectiveness and hpi?

```
SELECT
    NAME,
    CLUB,
    PERCENTAGE,
    HPI
FROM `lithe-aileron-433816-k5.handball.handball_stats_bundesliga`
WHERE TOTAL_GOALS > 30
ORDER BY HPI DESC
LIMIT 20;

SELECT
    CORR(PERCENTAGE, HPI) AS correlation_efficiency_hpi
FROM `lithe-aileron-433816-k5.handball.handball_stats_bundesliga`
WHERE TOTAL_GOALS > 40;
```



A correlation of **0.81** indicates a very strong positive relationship. In the scatter chart, the players align diagonally, proving that **shooting efficiency (Percentage)** is the dominant factor determining a player's final performance rating (**HPI**) in the Bundesliga.

- which players have >55%, more than 30 assists and less than 40 technical errors?

SELECT

NAME,
CLUB,
PERCENTAGE,
ASSISTS,
TURNOVERS,
HPI

FROM `lithe-aileron-433816-k5.handball.handball\_stats\_bundesliga`

WHERE PERCENTAGE > 55

AND ASSISTS > 30

AND TURNOVERS < 40

ORDER BY HPI DESC;

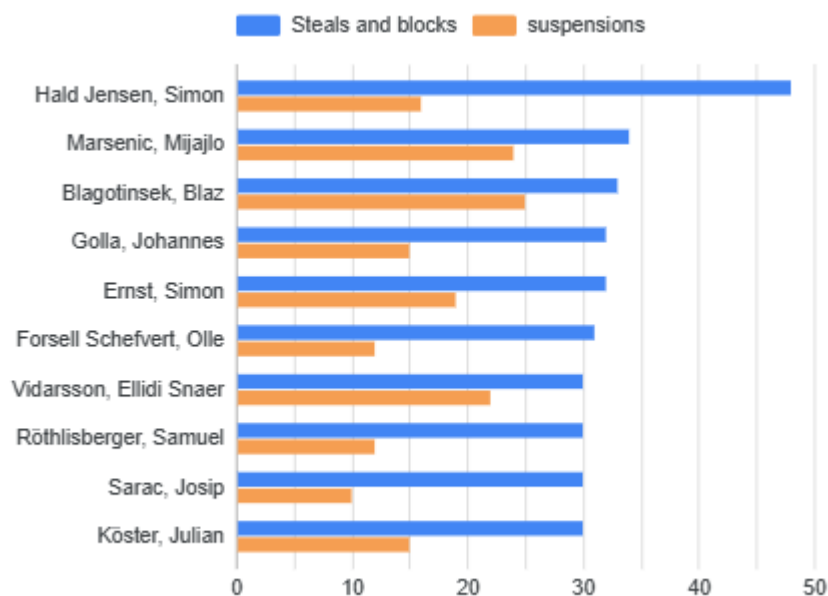
	NAME	TURNOVERS	HPI	ASSISTS	PERCENTAGE
1.	Magnusson, Omar Ingi	30	75	59	72
2.	Gidsel, Mathias	38	75	68	68
3.	Smits, Kay	29	74	44	72
4.	Mensing, Hans Aaron	13	74	42	60
5.	Gottfridsson, Jim	25	73	66	80
6.	Sagosen, Sander	13	73	41	64
7.	Steinert, Christoph	34	72	42	67
8.	Mappes, Dominik	28	72	67	60
9.	Johansson, Eric	30	72	55	62
10.	Andersson, Lasse Bredekjaer	28	72	59	56
11.	Bilyk, Nikola	27	71	36	59
12.	Rød, Magnus Abelvik	29	71	57	57
13.	Häfner, Kai	38	71	52	61
14.	Forsell Schefvert, Olle	29	71	55	75
15.	Duvnjak, Domagoj	14	71	44	66
16.	Kirkelokke, Niclas	33	70	54	62

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Insight: These players represent the "Elite Controllers." They manage a high workload (assists) with great efficiency (shooting %) while maintaining extreme discipline by minimizing technical errors (turnovers)

- which players have the biggest sum of steals and blocks compared the 2 minutes suspensions?

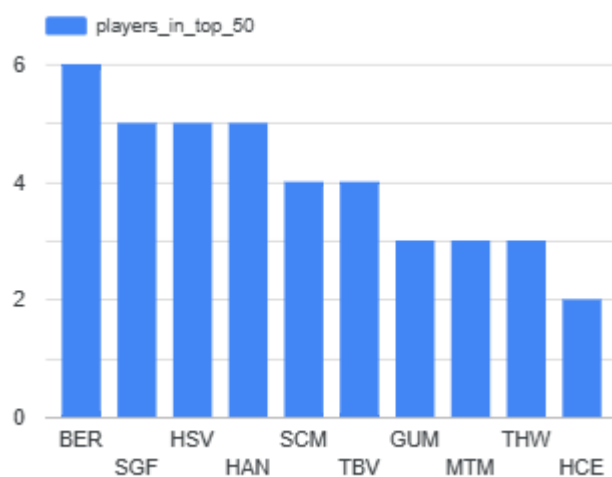
```
SELECT
    NAME,
    CLUB,
    STEALS,
    BLOCKS,
    `2MIN` AS suspensions,
    (STEALS + BLOCKS) AS total_defensive_actions,
    SAFE_DIVIDE((STEALS + BLOCKS), `2MIN`) AS clean_defense_ratio
FROM `lithe-aileron-433816-k5.handball.handball_stats_bundesliga`
WHERE `2MIN` > 5
    AND (STEALS + BLOCKS) > 10
ORDER BY clean_defense_ratio DESC
```



Insight: This "Clean Defense" metric identifies the most intelligent defenders. These players provide massive defensive value through steals and blocks without forcing their team into a numerical disadvantage via suspensions.

- Are the teams with the most players in the top 50 hpi in the top of the championship?

```
WITH Top50Players AS (  
  SELECT  
    NAME,  
    CLUB,  
    HPI  
  FROM `lithe-aileron-433816-k5.handball.handball_stats_bundesliga`  
  ORDER BY HPI DESC  
  LIMIT 50  
)  
SELECT  
  club,  
  COUNT(*) AS players_in_top_50,  
  ROUND(AVG(HPI), 1) AS average_hpi_of_top_players  
FROM Top50Players  
GROUP BY club  
ORDER BY players_in_top_50 DESC;
```



The top 5 teams of the championship were: **1. Kiel, 2. Magdeburg, 3. fuchse Berlin, 4. Flensburg, 5. Hannover**. This really shows that there is a correlation of the top teams and the top players with high hpi, except of the champions because as it is evident they had only 3 players in the top 50. This probably indicates that they won the championship and they were one step ahead of the other teams because they were not dependable by a few players but the results were a contribution of team work.

Act

Bundesliga Handball Analysis 2022-2023

Objective

This project analyzes player performance data from the Handball Bundesliga to identify the key metrics that drive success at both the individual and team levels. By leveraging SQL for data extraction and Looker Studio for visualization, the analysis aims to define what makes a player "valuable" in the modern game.

Key Findings

- **The Power of Efficiency:** A **very strong positive correlation (0.81)** was identified between shooting percentage and the Handball Performance Index (HPI). This confirms that high percentage finishing is the primary driver of a player's overall value rating.
- **Elite Creators:** Through the "Efficiency vs. Usage" analysis, the study isolated a rare group of players who maintain high shooting accuracy (>65%) while providing significant playmaking (Assists) with minimal turnovers.
- **Clean Defense:** The "**Clean Defense Ratio**" metric highlighted the league's most intelligent defenders—those who lead in steals and blocks while maintaining a low rate of 2-minute suspensions, thus keeping their teams at full strength.
- **Depth and Dominance:** The data shows a direct link between roster depth and league standings. Teams at the top of the championship consistently possess the highest density of "Top 50 HPI" players, proving that top-tier success requires multiple high-performing assets rather than a single star.

Conclusion

The analysis concludes that the "ideal" modern Bundesliga player is defined by **high-efficiency scoring, disciplined ball handling, and proactive, foul-free defense**. For clubs, the data suggests that recruiting for efficiency (Percentage) and defensive IQ (Ratio) is more predictive of success than simply looking at total goals scored and the use of roster depth helps the team bring results, plus the importance of the goalkeeper position is one of the most important factors.